MANCHESTER 1824
The University of Manchester

Understanding the Earth Workshop

Week 3 Stratigraphic correlation using absolute dating methods

The aims of this week's workshop are to:

- compare the subdivision of geologic time using relative and absolute dating methods
- introduce the geologic time scale and its major subdivisions
- introduce methods used to determine the absolute age of rock units including the concept of radioactive decay
- compare and contrast the most appropriate dating methods for different types and ages of material

1

MANCHESTER 1824
The University of Manchester

Understanding the Earth Workshop

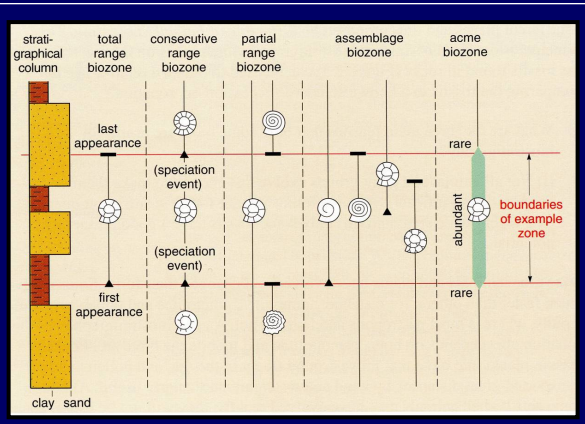
Week 3 Stratigraphic correlation using absolute dating methods

On successful completion of the workshop you should be able to:

- use lithostratigraphic and biostratigraphic methods to correlate stratigraphy across large areas
- recite the order of the major divisions and subdivisions of geologic time scale from the Earth's formation to the present day
- use ideas behind radiometric dating to date rock units
- select the most appropriate method to date a particular rock type or other material

2

Biostratigraphy – stratigraphic correlation using biozones



The fundamental unit of biostratigraphy is the biozone

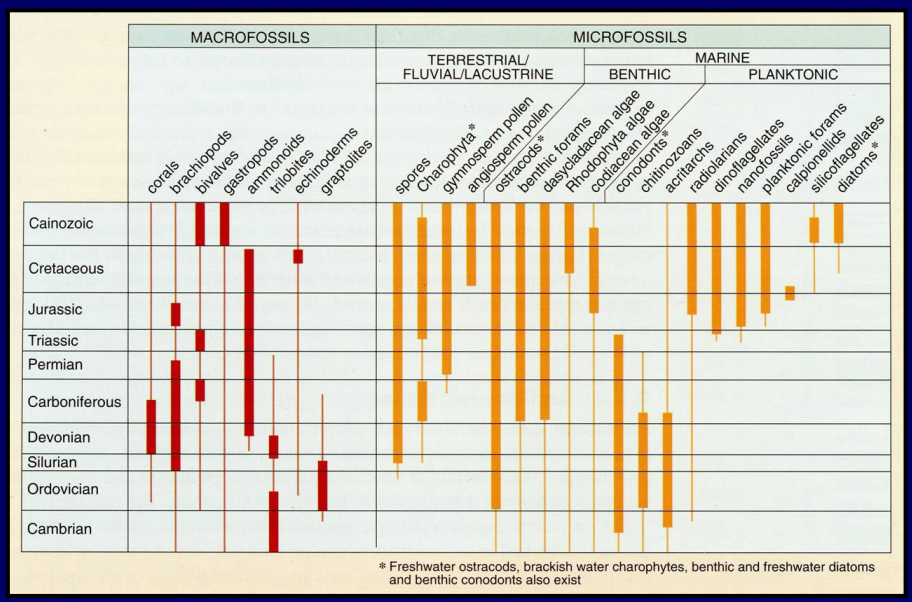
Biozones are defined by their zone fossils

The boundaries of the biozone may be specified in different ways

Total range biozone = biozone defined by the total (or local) range of one taxon
Consecutive range biozone = biozone defined by the range of one taxon, B, in the sequence A → B → C
Partial range biozone = biozone defined as that part of the range of fossil group B that is above the last appearance of fossil group A but below the first appearance of fossil group C
Assemblage biozone = biozone defined by the coexistence of a particular set of taxa
Acme biozone = biozone defined by the exceptional abundance of one taxon

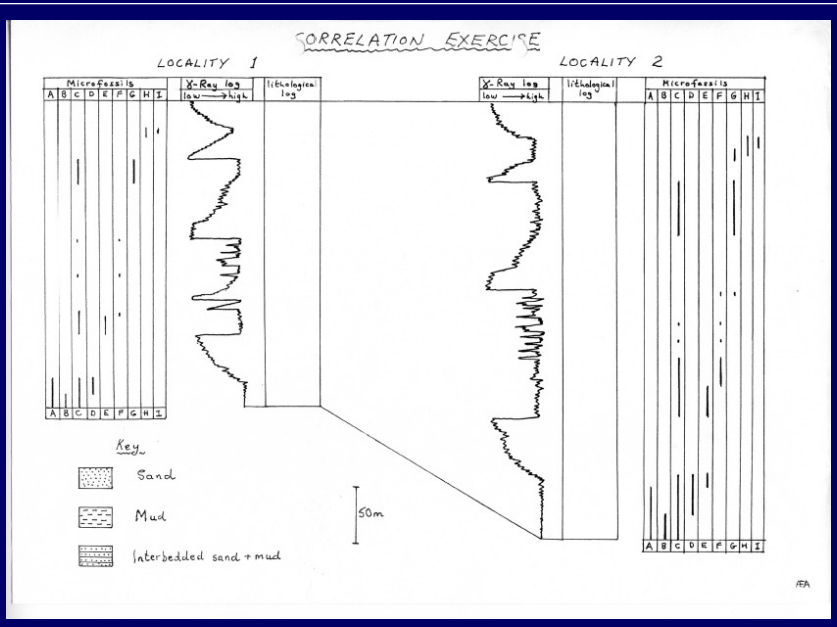
3

Fossils commonly used for biostratigraphic correlation



4

Correlation combining lithostratigraphy and biostratigraphy



5

Correlation combining lithostratigraphy and biostratigraphy

(a) Using the γ-ray log as a sand/mud indicator, complete the lithological logs at localities 1 and 2, using the symbols in the key.

(b) Using a coloured pen or pencil suggest lithological correlations between the two localities. The succession you are looking at lies between two volcanic ash bands which have been correlated, and are thus both lithological and time markers.

(c) Look at the distribution of microfossils at locality 2, and define some 'zones' based on assemblages of fossils. Which lithology contains the fossils? Why do fossils only occur in certain parts of the succession?

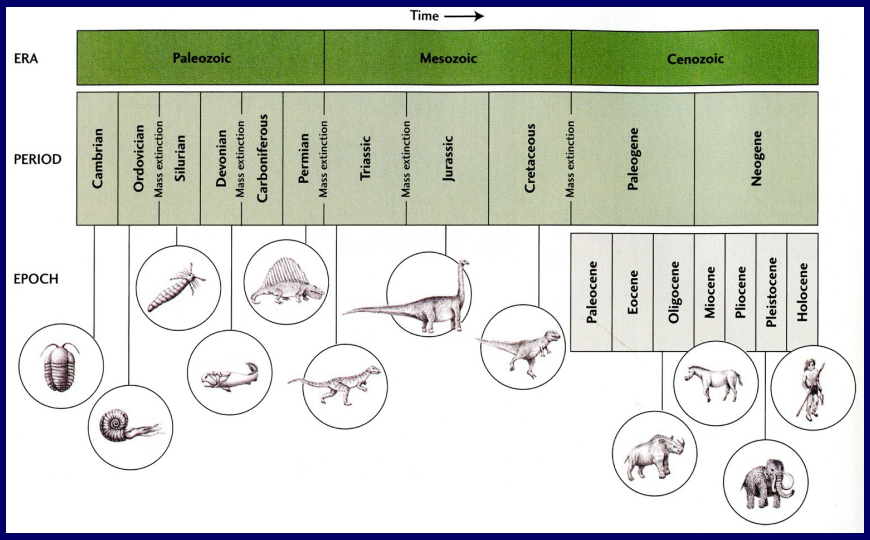
(d) Now look at the microfossil assemblages at Locality 1. Can you recognise the same zones you defined at Locality 2?

(e) Using a different coloured pen or pencil from that used for your lithostratigraphic correlations, suggest some tentative biostratigraphic correlations between the two localities.

(f) Comment briefly on the similarities and differences in the zones between the two localities and the significance of the correlations you have made.

6

The subdivision of time historically: chronostratigraphy



7

The division of geological time today





The golden spike defining the base of the Ediacaran Period (635 Ma), Enorama Creek, Flinders Ranges, South Australia

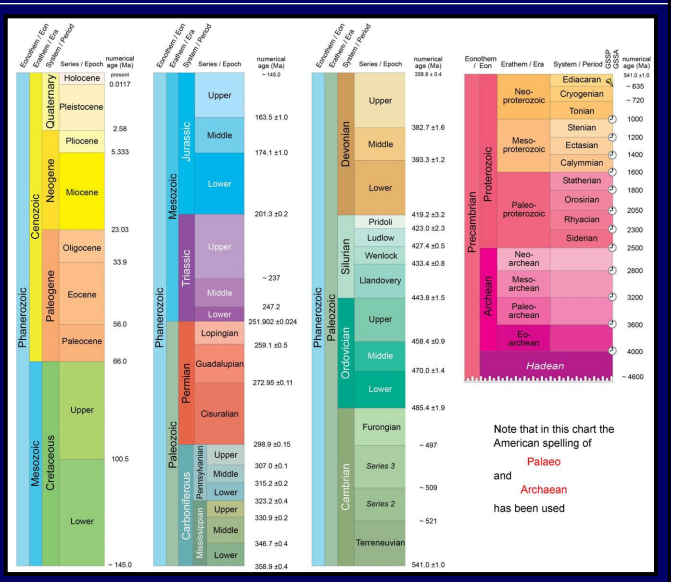
Global Stratotype Section and Point (GSSP) *aka a* "golden spike"

- define the base of geochronological units
- physical point within a succession of strata at a particular place somewhere in the world
- rocks at GSSP dated (i.e. located in an absolute time scale)

8

Subdivisions of geochronological units

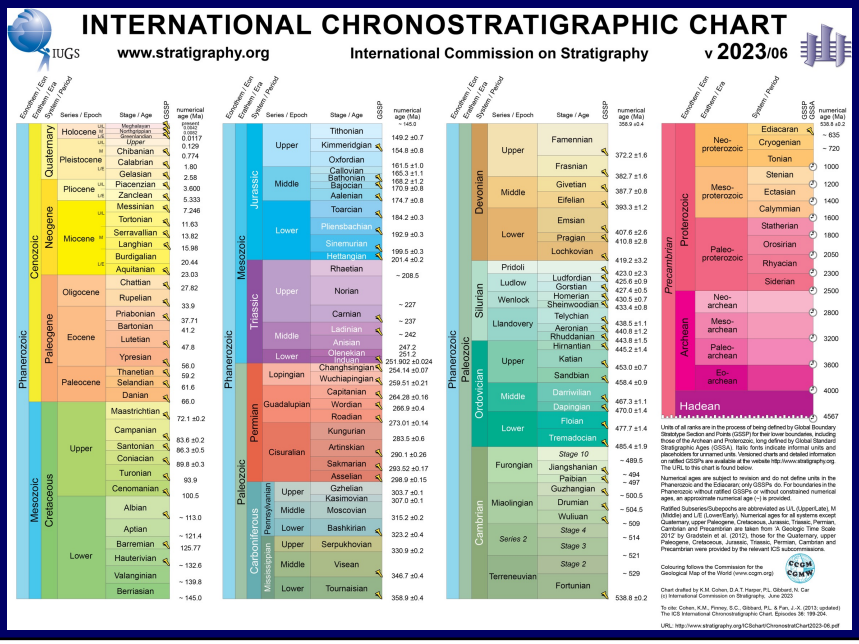
1. **Eons**
   Archaean, Proterozoic, Phanerozoic
   ↕ Precambrian
2. **Eras**
   (Precambrian eras), Palaeozoic, Mesozoic Cenozoic
3. **Periods**
   Cambrian, Ordovician, *etc.*
4. **Epochs**
   Some named, others Lower/Middle/Upper
5. **Ages**
   Smallest commonly used unit



www.stratigraphy.org

9

The 2023 geological timescale



10

Practice 1: Create your own memory aid

Create three mnenomics that will help you remember the order of the subdivision of geological time for the following time units:

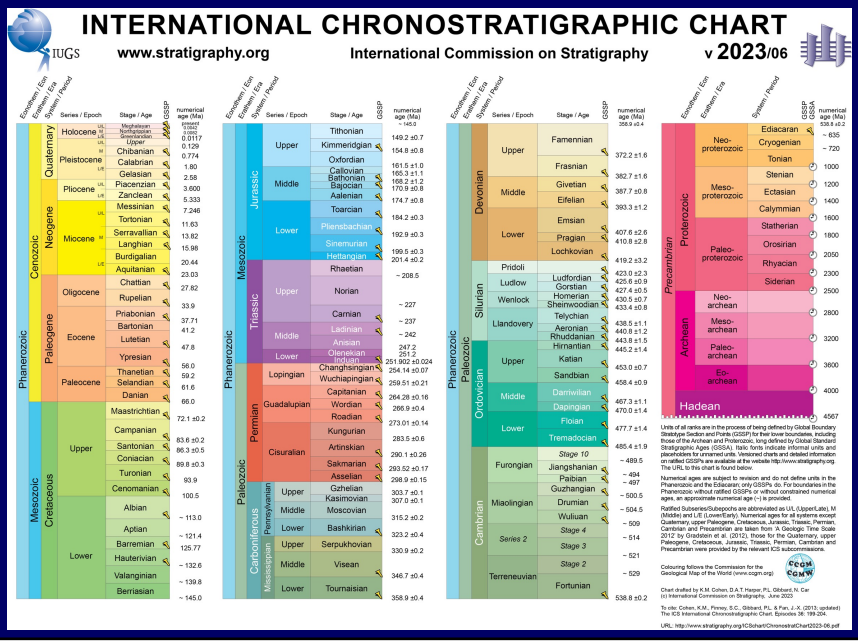
Order of time units in Eons

Palaeozoic, Mesozoic and Cenozoic: to the level of the Period

Cenozoic: to the level of the Epoch

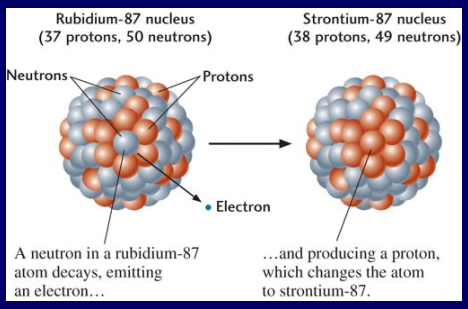
11

Absolute time



12

Absolute time and isotopic dating



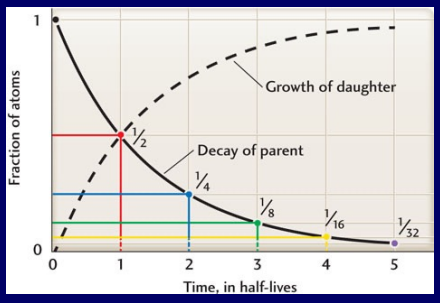
Radioactive decay equation

$$A = A_0 e^{-\lambda t}$$

λ is the decay constant for the isotope

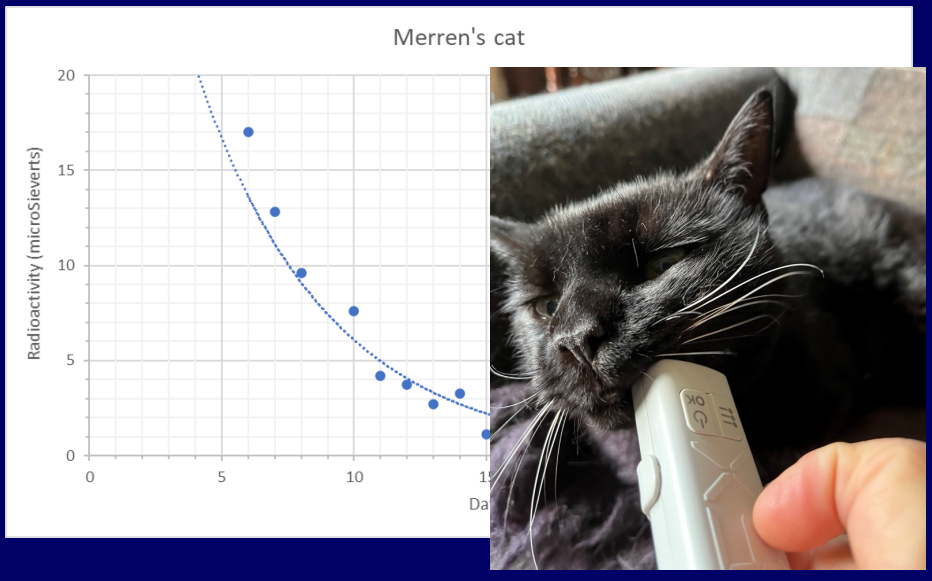
Half-life equation

$$t_{1/2} = \frac{ln2}{\lambda}$$



13

Practice 2: Merren's cat



14

Practice 3 Critical thinking: which method?

1. An Archeologist finds some cotton cloth at a burial site and wants to determine the age of the remains. Which isotopic system should they use? (^{14}C half life 5730 years; ^{238}U half-life 4468 Ma; ^{40}K half-life 1251 Ma)
   a. Carbon-14 b. Uranium c. Potassium-Argon

2. The Archeologist determines that there is 16.7% of the parent isotope remaining in the cloth sample. How old is the burial site? Hint: you can find the length of the half-life in the reading above.
   a. 13,559 years b. 14,798 years c. 16,743 years
   d. 1.66 billion years e. 1.81 billion years f. 2.05 billion

3. A geologist is trying to date a sequence of sedimentary rocks with abundant fossils and sandstones. Within the sequence is a distinctive clay layer that under closer inspection is fine-grained volcanic ash. Which of the following is the best way to obtain an absolute date for the sequence of rocks?
   a. Carbon date the fossils b. Potassium-Argon date the sands
   c. Uranium date the Zircons in the ash d. Identify the index fossils

4. The geologist determines there is 78.3% of the parent ^{238}U remaining in the sample that they examine. How old is the sequence of rocks? Hint: you can find the length of the half-life in the reading above. (^{238}U-^{206}Pb half-life 4468 million years)
   a. 187.5 million years b. 247.8 million years c. 390.7 million years
   d. 2.504 billion years e. 1.577 billion years f. 1.202 billion years

15

Can you do what we set out to do?

On successful completion of the workshop you will be able to:

- recite the order of the major divisions and subdivisions of geologic time scale from the Earth's formation to the present day
- use ideas behind radiometric dating to date rock units
- select the most appropriate method to date a particular rock type or other material

16